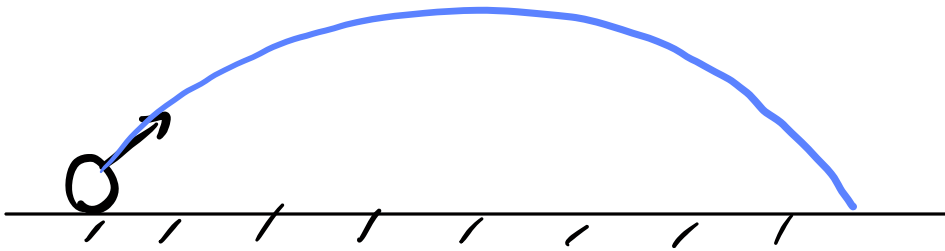


Projectile motion

Projection at an angle



u_x : Initial velocity (horizontal)

v_x : Horizontal velocity

u_y : Initial velocity (vertical)

v_y : Vertical velocity

CASE 1. Landing at the same height as at launch

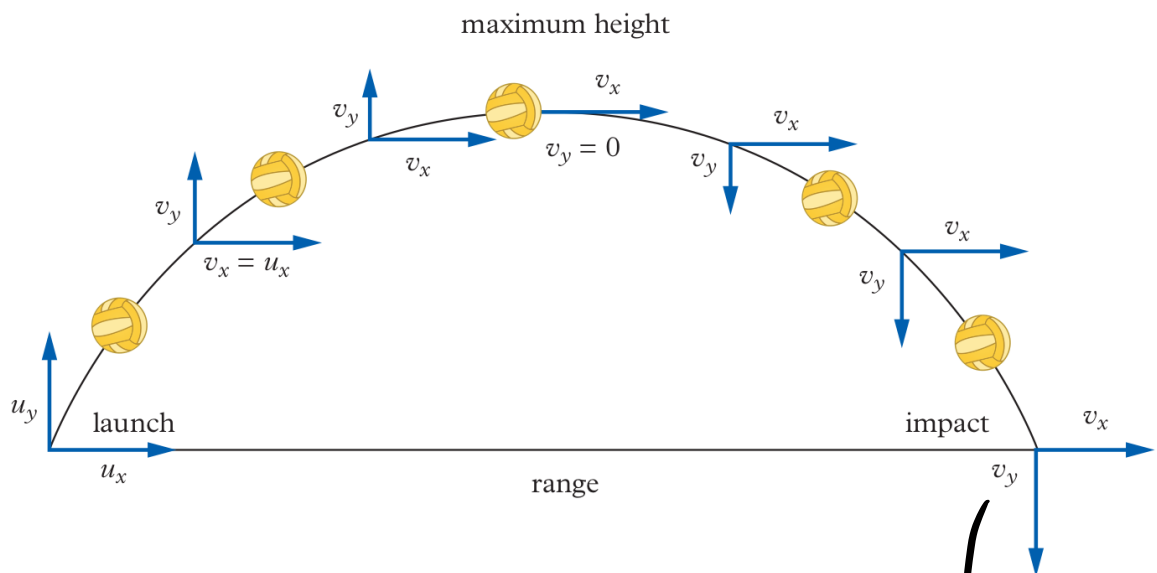


FIGURE 1 Horizontal and vertical components of velocity for the flight of a projectile

$$v_y = -u_y$$

* The horizontal component of velocity remains constant for the duration of the flight, hence $v_x = u_x$.

- u_x is the starting velocity in the horizontal (or x) direction.
- v_x is the final velocity in the horizontal (or x) direction.
- u_y is the starting velocity in the vertical (or y) direction.
- v_y is the final velocity in the vertical (or y) direction.
- g is the acceleration due to gravity (-9.8 m s^{-2}).
- s_x is the displacement in the horizontal (x) direction (also called the 'range').
- s_y is the displacement in the vertical (y) direction.

In horizontal direction, the acceleration (a) is zero.

$$v_x = u_x + at$$

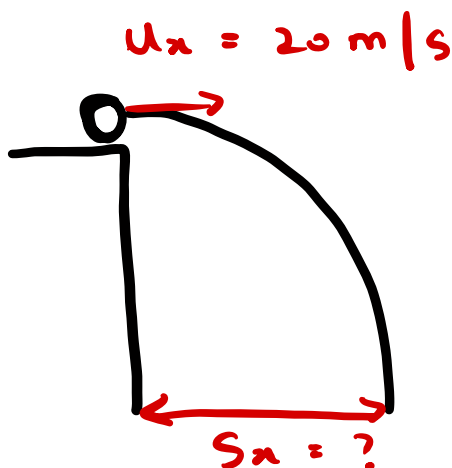
$$v_x = u_x$$

$$s_x = u_x t + \frac{1}{2} at^2$$

$$s_x = u_x t$$

WORKED EXAMPLE 1.2A

A golf ball is projected horizontally off a cliff at 20.0 m s^{-1} and it takes 4.0 s to reach the ground. Calculate how far out from the base of the cliff the ball will land.



$$u_x = 20 \text{ m/s}$$

$$t = 4 \text{ s}$$

$$s_x = u_x t = 20 \times 4 = 80 \text{ m}$$

* Vertical motion

In the vertical direction an object will accelerate downwards due to gravity, where the acceleration $a_y = -9.8 \text{ m/s}^2$.

$$v_y = u_y + a_y t$$

$$v_y^2 = u_y^2 + 2a_y s \quad \leftarrow \text{If no time given, then use this}$$

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$* v_y (\text{impact}) = -u_y (\text{launch})$$

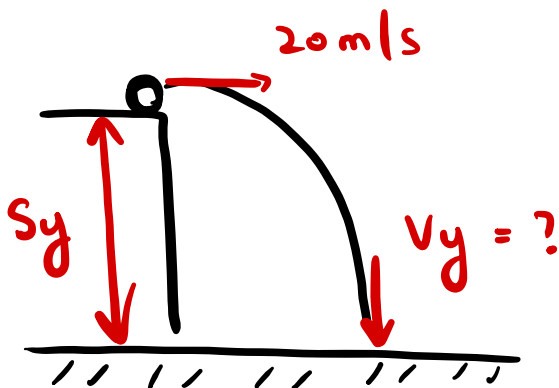
WORKED EXAMPLE 1.2A

A golf ball is projected horizontally off a cliff at 20.0 m s^{-1} and it takes 4.0 s to reach the ground. ~~Calculate how far out from the base of the cliff the ball will land~~

WORKED EXAMPLE 1.2B

For the golf ball in Worked example 1.2A, calculate:

- a the impact velocity in the vertical direction
- b the height of the cliff.



$$u_x = 20 \text{ m/s}$$

$$t = 4 \text{ s}$$

$$u_y = 0 \text{ m/s}$$

$$\begin{aligned} v_y &= u_y + a_y t = 0 + (-9.8) \times 4 \\ &= -39.2 \text{ m/s} \end{aligned}$$

$$s_y = u_y + \frac{1}{2} a_y t^2 = 0 + \frac{1}{2} (-9.8) \times 4^2$$

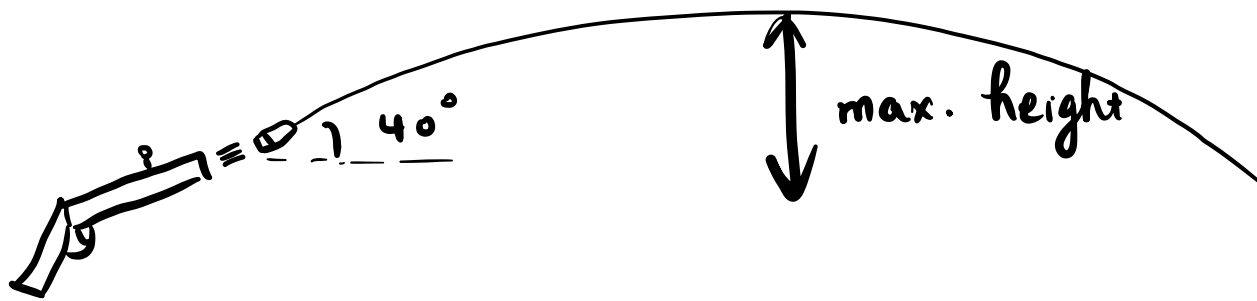
$$= -78.4 \text{ m}$$

Height of the cliff is 78.4 m.

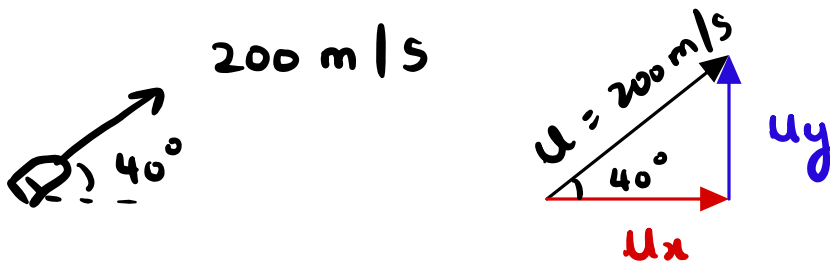
WORKED EXAMPLE 1.3A

A bullet was fired from a rifle at 200.0 m s^{-1} at an angle of 40.0° to the ground. Ignoring air resistance, calculate the:

- ✓ a initial vertical and horizontal components of the velocity
- ✓ b maximum height reached
- ✓ c time of flight (total time taken from start to finish)
- d horizontal range.



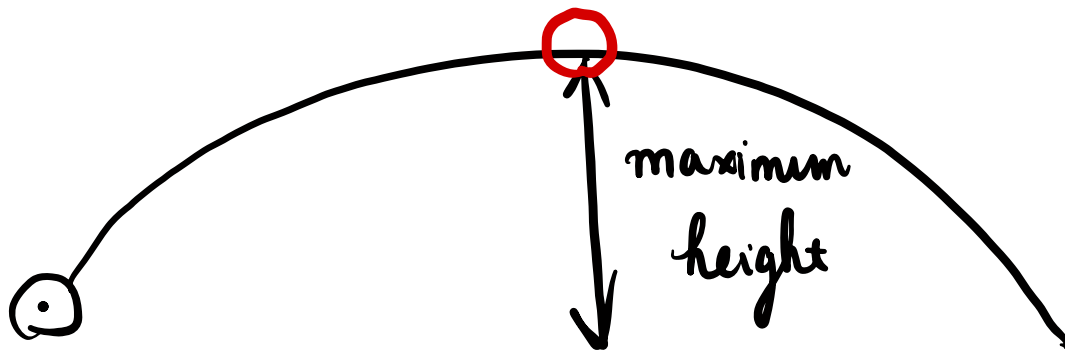
(a) u_x, u_y ?



$$u_x = u \cos \theta = 200 \cos 40^\circ = 153 \text{ m/s}$$

$$u_y = u \sin \theta = 200 \sin 40^\circ = 128.6 \text{ m/s}$$

(b)



At maximum height, $v_y = 0 \text{ m/s}$

$$u_y = 128.6 \text{ m/s}$$

$$v_y^2 = u_y^2 + 2a_y s_y$$

$$\Rightarrow s_y = \frac{v_y^2 - u_y^2}{2a_y} = \frac{0^2 - 128.6^2}{2 \times (-9.8)} = 844 \text{ m}$$

(c) 2 methods

① Determining time taken until the final vertical velocity is equal and opposite to initial vertical velocity.

$$v_y = -u_y = -128.6 \text{ m/s}$$

$$u_y = 128.6 \text{ m/s}$$

$$v_y = u_y + a_y t \Rightarrow t = \frac{v_y - u_y}{a_y}$$

$$t = \frac{-128.6 - 128.6}{-9.8}$$

$$t = 26.24 \text{ s}$$

② Determining the time until $s_y = 0$ again.

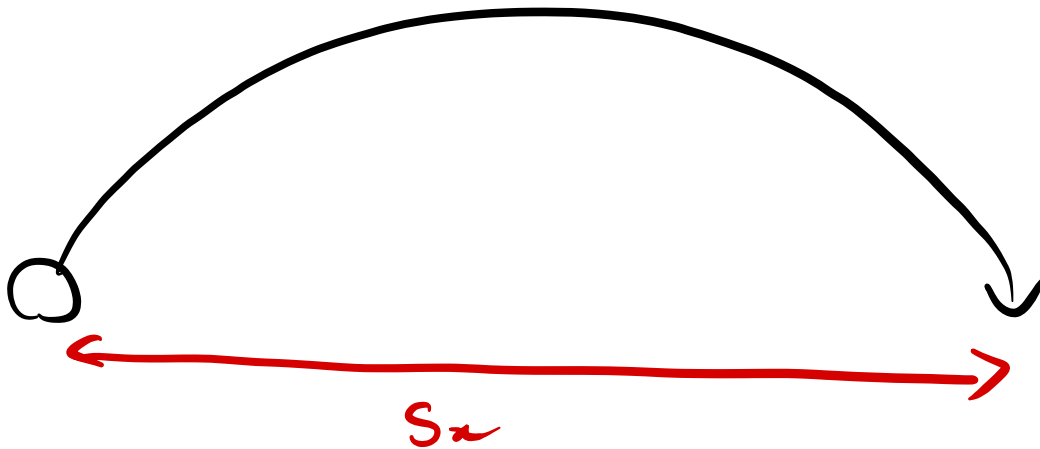
$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$0 = 128.6 t + \frac{1}{2} (-9.8) t^2$$

$$128.6 \cancel{t} = 4.9 t \cancel{t}$$

$$t = 26.24 \text{ s}$$

(d)



$$\begin{aligned} S_x &= u_x t + \cancel{\frac{1}{2} a_x t^2}^0 \\ &= 153 \times 26.24 \\ &= 4020 \text{ m} \end{aligned}$$

CASE 2. Landing at a lower level than at launch.

Landing at a lower height than at launch is similar to a person launching a shot-put at shoulder height and it hitting the ground at foot level, or a skier launching upwards off a ramp and landing further down the mountain (as in Figure 2). The main difference between this and Case 1, landing at the same height as at launch, is that the vertical displacement at impact is no longer zero but some negative quantity.



FIGURE 2 A skier jumping from a bump will land at a lower height than that from which they took off.

WORKED EXAMPLE 1.3B

A cannon is fired from the edge of a cliff, which is 60.0 m above the sea (Figure 3).

The initial velocity of the cannonball is 88.3 m s^{-1} and it is fired at an upwards angle of 34.5° to the horizontal. Determine:

- a the time the ball is in the air t
- b the impact velocity v
- c the horizontal distance from the base of the cliff that the ball strikes the water. s_x

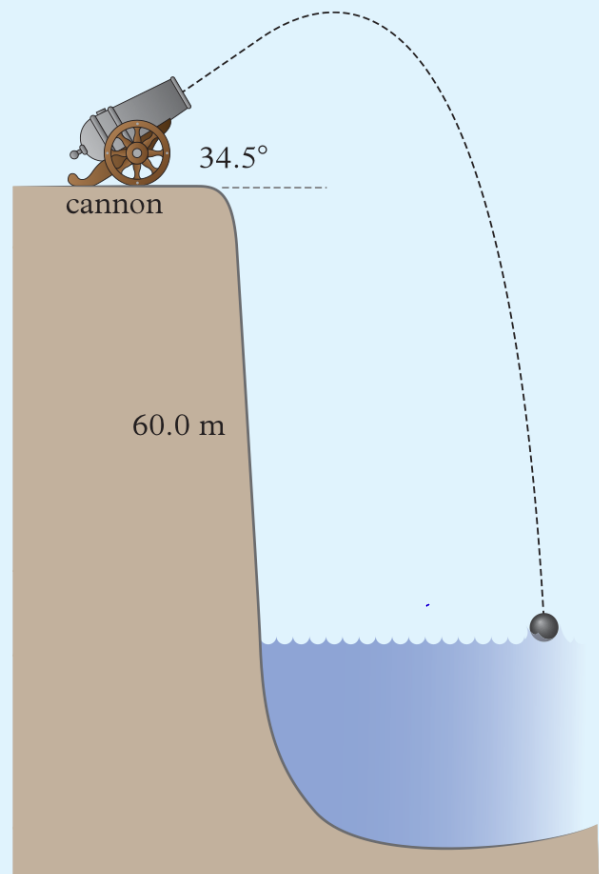


FIGURE 3 The projectile lands lower than launch.

$$u = 88.3 \text{ m/s} \quad \theta = 34.5^\circ$$

$$u_x = u \cos \theta = 88.3 \cos 34.5^\circ = 72.8 \text{ m/s}$$

$$u_y = u \sin \theta = 88.3 \sin 34.5^\circ = +50 \text{ m/s} \quad (\uparrow)$$

$$s_y = -60 \text{ m}$$

$$a_y = -9.8 \text{ m/s}^2$$

(a) Find t ,

$$\text{Method 1. } s_y = u_y t + \frac{1}{2} a_y t^2$$

$$-60 = 50t + \frac{1}{2}(-9.8)t^2$$

$$-4.9t^2 + 50t + 60 = 0$$

$$t = -1.08 \text{ s} \quad \text{or} \quad 11.3 \text{ s}$$

$$t = 11.03 \text{ s}$$

Method 2. Find v_y first ($v_y^2 = u_y^2 + 2a_y s_y$)
 then t ($v_y = u_y + a_y t$)

$$v_y^2 = 50^2 + 2 \times (-9.8) \times (-60)$$

$$\Rightarrow v_y = \pm 60.6 \text{ m/s}$$

$\therefore$ Ball is moving down

$$\therefore v_y = -60.6 \text{ m/s}$$

$$v_y = u_y + a_y t$$

$$-60.6 = 50 + (-9.8)t$$

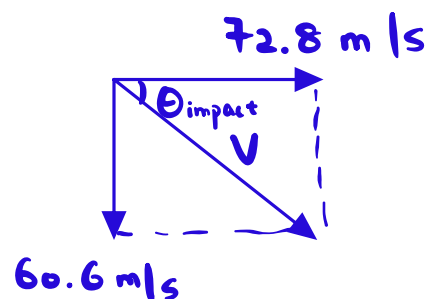
$$t = 11.3 \text{ s}$$

(b) ① v

② θ_{impact}

$$\cdot v_x = u_x = 72.8 \text{ m/s}$$

$$\cdot v_y = -60.6 \text{ m/s}$$



$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(72.8)^2 + (-60.6)^2} = 94.7 \text{ m/s}$$

$$\begin{aligned}\theta_{\text{impact}} &= \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{60.6}{72.8}\right) \\ &= 39.8^\circ \text{ to the horizontal}\end{aligned}$$

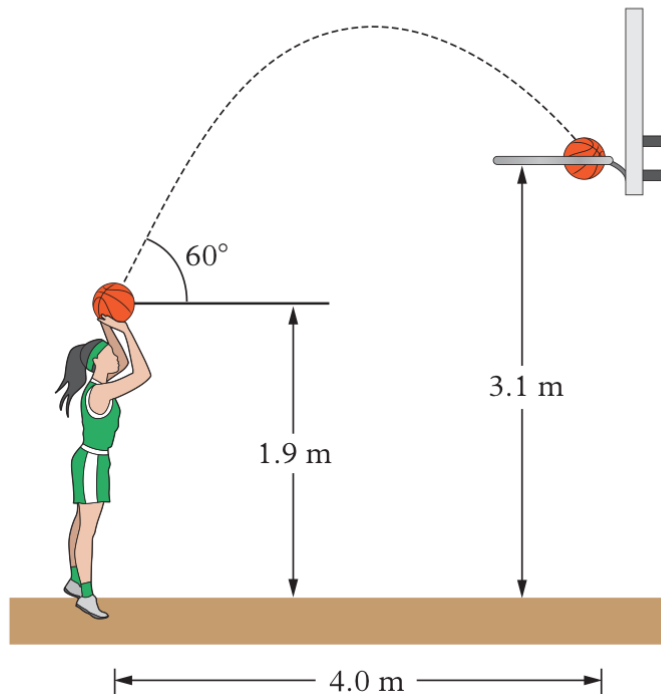
$$(c) \quad S_x = u_x t = 72.8 \times 11.3 = 823 \text{ m}$$

CASE 3. Landing at higher than launch height

The final variation on projectile trajectory is where it lands higher than at launch height. A good example of this is a basketball player shooting a ball into a hoop.

For example, in Figure 6 the ball leaves the player's hand at an initial vertical displacement of +1.9 m and lands in the hoop at a final vertical displacement of +3.1 m. You can simplify this by saying the initial vertical displacement is 0 m (where it leaves the player's hands), and the final vertical displacement at the hoop is +1.2 m (higher than at the start). For this example, $s_y = +1.2$ m, s_x (range) = 4.0 m, $\theta = 60^\circ$ upwards, $g = -9.8$ m s⁻².

There is a question like this in Check your learning 1.3.



QUESTION 19 (5 marks)

Consider the relation $x = \ln(e^{-x} + e^{-y})$.

Show that $\frac{dy}{dx} < -1$ for all points on the relation.

$$x = \ln(e^{-x} + e^{-y})$$

$$e^x = e^{\ln(e^{-x} + e^{-y})}$$

$$e^x = e^{-x} + e^{-y}, \text{ Find } \frac{dy}{dx}$$

Implicit differentiation

$$\Rightarrow e^x = -e^{-x} + \frac{dy}{dx}(-e^{-y})$$

$$\frac{dy}{dx} \cdot e^{-y} = -e^{-x} - e^x$$

$$\frac{dy}{dx} = \frac{-e^{-x} - e^x}{e^{-y}} = \frac{-e^{-x} - e^x}{e^x - e^{-x}} = \frac{-(e^{-x} + e^x)}{e^x - e^{-x}}$$

$\therefore$ For any $y \in \mathbb{R}$, $e^{-y} > 0$.

$$e^x - e^{-x} > 0$$

$$e^x > e^{-x} \Rightarrow x > 0$$

For $x > 0$:

$$e^{-x} > 0, \quad e^x + e^{-x} > e^x - e^{-x}$$

$$\frac{e^x + e^{-x}}{e^x - e^{-x}} > \frac{e^x - e^{-x}}{e^x - e^{-x}}$$

$$\frac{e^x + e^{-x}}{e^x - e^{-x}} > 1$$

$$\frac{-(e^x + e^{-x})}{e^x - e^{-x}} < -1$$

Recall $\frac{dy}{dx} = \frac{-(e^{-x} + e^x)}{e^x - e^{-x}}$

$$\Rightarrow \frac{dy}{dx} < -1$$

Q.E.D.